betaMC: Staging

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1 Monte Carlo Simulation

```
# Fit the regression model
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
# Generate the sampling distribution of parameter estimates
mc <- MC(object, R = 20000, type = "mvn", seed = 42)

#> Error in `.MCScreenLM()`:
#> !  object '_betaMC_MCScreenLM' not found
```

2 Standardized Regression Slopes

```
out <- BetaMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found

coef(out)

#> Error:
#> !  object 'out' not found

vcov(out)

#> Error:
#> !  object 'out' not found
```

```
confint(out)
```

```
#> Error:
#> !  object 'out' not found
```

3 Multiple Correlation

```
out <- RSqMC(mc)
```

```
#> Error:
#> !  object 'mc' not found
```

```
# Methods ------------------------------------------------------------
print(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
summary(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
coef(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
vcov(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
confint(out)
```

```
#> Error:
#> !  object 'out' not found
```

4 Semipartial Correlation

```
out <- SCorMC(mc)
```

```
#> Error:
#> !  object 'mc' not found
```

```
# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found

coef(out)

#> Error:
#> !  object 'out' not found

vcov(out)

#> Error:
#> !  object 'out' not found

confint(out)

#> Error:
#> !  object 'out' not found
```

5 Improvement in R-Squared

```
out <- DeltaRSqMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found

coef(out)

#> Error:
#> !  object 'out' not found
```

```
vcov(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
confint(out)
```

```
#> Error:
#> !  object 'out' not found
```

6 Squared Partial Correlation

```
out <- PCorMC(mc)
```

```
#> Error:
#> !  object 'mc' not found
```

```
# Methods ------------------------------------------------------
print(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
summary(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
coef(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
vcov(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
confint(out)
```

```
#> Error:
#> !  object 'out' not found
```

7 Differences of Standardized Slopes

```r
out <- DiffBetaMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found

coef(out)

#> Error:
#> !  object 'out' not found

vcov(out)

#> Error:
#> !  object 'out' not found

confint(out)

#> Error:
#> !  object 'out' not found
```

8 Monte Carlo Simulation - Multiple Imputation

```r
nas1982_missing <- mice::ampute(nas1982, mech = "MCAR")$amp
# Fit the regression model
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982_missing)
# Generate the sampling distribution of parameter estimates
mi <- mice::mice(nas1982_missing, m = 100, seed = 42, print = FALSE)
mc <- MCMI(object, mi = mi, R = 20000, type = "mvn",
           seed = 42)

#> Error in `.MCScreenLM()`:
#> !  object '_betaMC_MCScreenLM' not found
```

9 Standardized Regression Slopes

```
out <- BetaMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found

coef(out)

#> Error:
#> !  object 'out' not found

vcov(out)

#> Error:
#> !  object 'out' not found

confint(out)

#> Error:
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```

10 Multiple Correlation

```
out <- RSqMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)

#> Error:
#> !  object 'out' not found
```

```
coef(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
vcov(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
confint(out)
```

```
#> Error:
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```

11 Semipartial Correlation

```
out <- SCorMC(mc)
```

```
#> Error:
#> !  object 'mc' not found
```

```
# Methods ------------------------------------------------------------
print(out)
```

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summary(out)
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confint(out)
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12 Improvement in R-Squared

```
out <- DeltaRSqMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

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summary(out)

#> Error:
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coef(out)

#> Error:
#> !  object 'out' not found

vcov(out)

#> Error:
#> !  object 'out' not found

confint(out)

#> Error:
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```

13 Squared Partial Correlation

```
out <- PCorMC(mc)

#> Error:
#> !  object 'mc' not found

# Methods ------------------------------------------------------------
print(out)

#> Error:
#> !  object 'out' not found

summary(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
coef(out)
```

```
#> Error:
#> !  object 'out' not found
```

```
vcov(out)
```

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#> Error:
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```

```
confint(out)
```

```
#> Error:
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```

14 Differences of Standardized Slopes

```
out <- DiffBetaMC(mc)
```

```
#> Error:
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```
# Methods ------------------------------------------------------
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```
summary(out)
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coef(out)
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vcov(out)
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#> Error:
#> !  object 'out' not found
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```
confint(out)
```

```
#> Error:
#> !  object 'out' not found
```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/